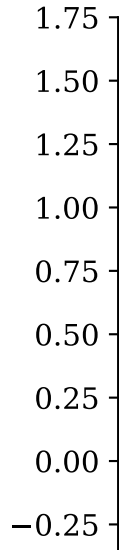


Each transverse exchange costs one power of $J_{\pm}$ and climbs J_{zz} ;
one extra rung makes the hexagon a weaker, higher-order process

Four-loop (2nd order)

$$g_4 = \frac{4J_{\pm}^2}{J_{zz}}$$

virtual energy above the ice manifold



Hexagon (3rd order)

$$g_{\text{hex}} = \frac{12|J_{\pm}|^3}{J_{zz}^2}$$

